



Brazil can trigger a tornado in Texas

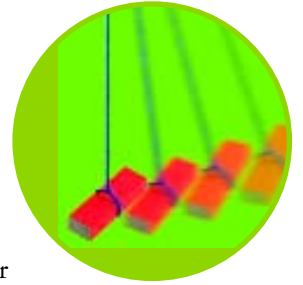
Make a chaos pendulum

A pendulum is a weight that swings back and forth on a length of string. Ordinary pendulums are predictable, but you can make a totally unpredictable "chaos pendulum"

by using magnets. Use a bar magnet as the weight and suspend this over a table. Fix three more

magnets to the table below, making sure the pendulum can't touch them.

Swing the weight and watch what happens.



START A HURRICANE

Weather works a bit like pinball. When forecasters try to predict the weather, they find that tiny differences in the starting conditions lead to totally different outcomes after 4–5 days. We call it the butterfly effect, because it means that a butterfly beating its wings in Brazil could, in theory, cause a tornado in Texas. You can do exactly the same thing – if you blink you might cause a hurricane in Hawaii or a typhoon in Taiwan!



Chaos in the bathroom

You can see chaos with your own eyes by slowly turning on a tap. First it drips. Open it a bit more. Now a smooth, steady trickle comes out. Open it a bit more and the trickle gets messy – it gurgles, twists, and splashes. The flow has become "turbulent", which means its motion is chaotic. A similar thing happens when you put a candle out. A smooth stream of smoke rises predictably for a few inches, then it suddenly turns chaotic, swirling and rippling in complicated patterns.



other planets through space is as chaotic as pinball